Yingqiang Wang

Ph.D. | Postdoctoral Research Fellow

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Google Scholar, 309 Citations, 10 h-index

PROFILE

Y. Wang, born in 1997, is currently a Postdoc at SJTU. He received his B.Eng. (2019) and Ph.D. (2024) degrees from Zhejiang University and was a joint Ph.D. at ETH Zurich from 2022 to 2023. His research centers on multimodal acoustic sensing, positioning, and navigation for underwater intelligent systems. He developed a flexible-waveguide acoustic sensing method featured on the cover of Nature Electronics, and the piUSBL acoustic system proved successful in sea trials. As first/corresponding author, he has published 9 papers in top journals including **Nature Electronics** (cover article), **IEEE JOE**, **IEEE RA-L**, and **IEEE JSEN**, with a total of 19 indexed publications.

Beyond exploring new theories and mechanisms, he places particular emphasis on the development of embedded hardware/software systems, as well as field experiments for rigorous validation.

ACADEMIC APPOINTMENTS

- Oct 2024 – Now — Postdoc Research Fellow
  Shanghai Jiao Tong University — School of Oceanography

EDUCATION

- Nov 2022 – Nov 2023 — Joint Ph.D. (Advisor: Prof. Daniel Ahmed)
  ETH Zurich, Switzerland — Institute of Robotics and Intelligent Systems
- Sep 2019 – Sep 2024 — Ph.D. (Supervisor: Prof. Ying Chen)
  Zhejiang University, China — Institute of Ocean Technology
- Sep 2015 – Jul 2019 — B.Eng. (Supervisor: Prof. Ying Chen)
  Zhejiang University, China — Institute of Ocean Technology

RESEARCH GRANTS

- **National Natural Science Foundation of China** — Young Scholars Program — [52501422]
  Project Leader [2026-2028] RMB 300,000
  Topic: Research on the Mechanisms of Underwater Positioning and Sensing Using a Flexible Ultra-Short Baseline Acoustic Array
- **Shanghai Natural Science Foundation** — Young Scholars Program — [25ZR1402253]
  Project Leader [2025-2028] RMB 250,000
  Topic: Acoustic Positioning and Sensing for Marine Robotics Using Flexible USBL Arrays
- **Zhejiang University Academic Award for Outstanding PhD Candidates** — [2022052]
  Project Leader [2022-2024] RMB 20,000
  Topic: Integrated Navigation for AUVs based on Dual-Beacon piUSBL

- **National Key R&D Program of China** [2017YFC0306100]
  Participant [2018-2022] RMB 7,800,000
  Topic: Acoustic Navigation System for the Autonomous Underwater Helicopter (*Student leader*)

JOURNAL PUBLICATIONS

[1] **Y. Wang**, C. Sun, D. Ahmed, A smart acoustic textile for health monitoring. Nature Electronics 8, 485–495 (2025). (*Cover Article, Editorial, News & Views, ETHz News, Nature Portfolio*)

[2] **Y. Wang**, R. Hu, P. Du, W. Yang, Y. Chen, S. H. Huang, One-Way-Travel-Time Hybrid Baseline Navigation for Micro Autonomous Underwater Vehicles. IEEE Journal of Oceanic Engineering 50, 968–984 (2025).

[3] **Y. Wang**, R. Hu, Y. Chen, S. H. Huang, Adaptive noise cancelling for an AUV-mounted passive inverted USBL array. Ocean Engineering 288, 115998 (2023).

[4] **Y. Wang**, R. Hu, S. H. Huang, Z. Wang, P. Du, W. Yang, Y. Chen, Passive Inverted Ultra-Short Baseline Positioning for a Disc-Shaped Autonomous Underwater Vehicle: Design and Field Experiments. IEEE Robotics and Automation Letters 7, 6942–6949 (2022). (ICRA 2023 Oral)

[5] **Y. Wang**, S. H. Huang, Z. Wang, R. Hu, M. Feng, P. Du, W. Yang, Y. Chen, Design and experimental results of passive iUSBL for small AUV navigation. Ocean Engineering 248, 110812 (2022).

[6] **Y. Wang**, S. H. Huang, M. Feng, Y. Chen, Simultaneous Subsea Gas Plumes Detection and Positioning Using a Transceiver of an Ultra-Short Baseline System. IEEE Sensors Journal 22, 5778–5786 (2022).

[7] J. Huang, H. Li, Z. Liu, Z. Wang, **Y. Wang***, Y. Chen*, GNSS-aided installation error compensation for DVL/INS integrated navigation system using error-state Kalman filter. Measurement 242, 116224 (2025).

[8] X. Song, Y. Bai, Y. Bi, **Y. Wang***, Z. Zeng*, H. Zhang, F. Zhou, L. Lian, Nezha-H: An HAUV for Aerial and Underwater Observation and Sampling. IEEE Robotics and Automation Letters 10, 6776–6783 (2025).

[9] J. Huang, **Y. Wang***, H. Li, Z. Liu, Z. Wang, Y. Chen*, Precise Time Delay Measurement and Compensation for Tightly Coupled Underwater SINS/piUSBL Navigation. IEEE Transactions on Instrumentation and Measurement, early access (2026), doi: 10.1109/TIM.2026.3676179.

[10] J. Huang, **Y. Wang***, Y. Chen*, Raspi^{2}USBL: An open-source Raspberry Pi-Based Passive Inverted Ultra-Short Baseline Positioning System for Underwater Robotics. IEEE Journal of Oceanic Engineering, in revision

[11] K. Han, Y. Bi, Y. Bai, **Y. Wang***, H. Zhou, Z. Zhang, Z. Zeng*, L. Lian, Robust Submergence and Wave-Adaptive Cross-Domain Strategy for Hybrid Aerial Underwater Vehicles in Challenging Sea States. IEEE Transactions on Field Robotics, in revision

[12] C. Sun, Z. Li, A. Osman, **Y. Wang**, H. Yu, W. Wang, Z. Yu, Y. He, Y. Chen, X. Chen, Z. Mao, Y. Li, Y. Liu, M. Miao, Y. Wang, D. Ahmed, J. Lu, Reconfigurable Acoustic Printing for Conformal and Minimally Invasive Repair. Advanced Functional Materials, e25723 (2025).

[13] P. Du, **Y. Wang**, W. Yang, R. Hu, Y. Chen, S. H. Huang, A hovering micro autonomous underwater vehicle with integrated control and positioning system. Ocean Engineering 304, 117819 (2024).

[14] R. Hu, **Y. Wang**, Y. Chen, S. H. Huang, Fast Calibration of Superdirective Ultra-short Baseline Array. Journal of Marine Science and Engineering 11, 1665 (2023)

[15] J. Zhou, H. Huang, S. H. Huang, Y. Si, K. Shi, X. Quan, C. Guo, C.-W. Chen, Z. Wang, **Y. Wang**, Z. Wang, C. Cai, R. Hu, Z. Rong, J. He, M. Liu, Y. Chen*, AUH, a New Technology for Ocean Exploration. Engineering 25, 21–27 (2023).

[16] C. Cai, Z. Rong, Z. Chen, B. Xu, Z. Wang, S. Hu, **Y. Wang**, M. Dong, X. Quan, Y. Si, Y. Chen, H. Huang, A resident subsea docking system with a real-time communication buoy moored by an electro-optical-mechanical cable. Ocean Engineering 271, 113729 (2023).

[17] P. Du, W. Yang, **Y. Wang**, R. Hu, Y. Chen, S. H. Huang, A novel adaptive backstepping sliding mode control for a lightweight autonomous underwater vehicle with input saturation. Ocean Engineering 263, 112362 (2022).

[18] P. Du, S. H. Huang, W. Yang, **Y. Wang**, Z. Wang, R. Hu, Y. Chen, Design of a Disc-Shaped Autonomous Underwater Helicopter with Stable Fins. Journal of Marine Science and Engineering 10, 67 (2022).

CONFERENCE PROCEEDINGS

[1] M. Aili, X. Song, **Y. Wang***, Z. Zeng*, L. Lian, "Nezha-Morphing: Design and Experiments of a Seabird-Inspired Hybrid Aerial Underwater Vehicle" in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (Hangzhou, China, 2025), pp. 3422–3427.

[2] W. C. Yang, P. Z. Du, R. Y. Hu, **Y. Wang**, D. W. Xu, S. H. Huang, Adaptive Backstepping Sliding Mode Attitude Control for AUVs Subject to Model Errors and Unknown Disturbances, in *OCEANS 2022, Hampton Roads* (Hampton Roads, VA, USA, 2022), pp. 1–5.

[3] M. Y. Feng, S. H. Huang, T. C. Yang, Q. Y. Tu, X. H. Wan, **Y. Wang**, Detection of Gas Leaks from Sea Bed Using a Small Circular Array, in *OCEANS 2019 - Marseille* (Marseille, France, 2019), pp. 1–5.

INVITED TALKS

[1] July 2025 "Nezha-HAUVs: Hybrid Aerial Underwater Vehicles for Marine Observation and Sampling", *the Opening Ceremony of 2025 Global Summer School, Shanghai Jiao Tong University,* Shanghai, China.

[2] May 2023 "Passive Inverted Ultra-Short Baseline Positioning for a Disc-shaped Autonomous Underwater Vehicle: Design and Field Experiments," *the 40th IEEE International Conference on Robotics and Automation (ICRA 2023),* London, UK.

[3] May 2023 "Passive Inverted Ultra-Short Baseline Positioning for Autonomous Underwater Vehicles," *the 6th China Ocean Technology Conference*, Zhoushan, China.

[4] July 2022 "Passive Inverted Ultra-Short Baseline Positioning for Small AUV Navigation," *the Youth Academic Salon of Ocean College, Zhejiang University*, Zhoushan, China.

CRUISE PARTICIPATION

- Aug 2021 R/V *YueZhanYuKe-3* The South China Sea
  Sea trials of the Autonomous Underwater Helicopter
- Jul 2021 R/V *YueZhanYuKe-3* The South China Sea
  Sea trials of the Autonomous Underwater Helicopter
- Apr-Jun 2021 R/V *RuiLi-11* Qiandao Lake, Hangzhou
  Field lake trials of the Autonomous Underwater Helicopter

TECHNICAL EXPERTISE

- Digital Signal Processing (MATLAB, C)
- Embedded System Development (DSP, ARM)
- Electrical Design (Altium Designer)
- Mechanical Design (SOLIDWORKS)

ACADEMIC ACTIVITIES

- Membership
  ASC (The Acoustical Society of China) Member
  IEEE Member (Oceanic Engineering Society, Robotics and Automation Society)
- Reviewer
  IEEE Journal of Oceanic Engineering (JOE)
  The Journal of the Acoustical Society of America (JASA)
  IEEE International Conference on Robotics and Automation (ICRA)
  IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
  Telecommunication Systems

CAMPUS ACTIVITIES

- 2024.10 – now **Athlete**
  Dragon Boat Team of Faculty and Staff, Shanghai Jiao Tong University
- Sep 2019 – Dec 2021 **Founding Coach**
  Water Sports Team of Ocean College, Zhejiang University
- Sep 2015 – Jul 2019 **Athlete**
  Water Sports Team of Zhejiang University

AWARDS AND HONORS

2025	Postdoctoral Fellowship Program of China Postdoctoral Science Foundation	
2025	Shanghai Post-doctoral Excellence Program	
2024	Zhejiang University Academic Award for Outstanding Doctoral Candidates	
2022	China Scholarship Council (CSC) Scholarship	
2022	Outstanding Graduate Student of Zhejiang University	
2019	Outstanding Graduate of Zhejiang Province & Zhejiang University	
2018	TOP 10 Students of Ocean College, Zhejiang University	
2023	Swiss Dragon Boat Championships (Open 200 m, DBC Eglisau)	Gold Medalist
2021	The 5th National University Canoe Championships (Men's K1)	Gold Medalist
2021	The 5th National University Canoe Championships (Men's K2)	Gold Medalist
2018	The 3rd National University Canoe Championships (Men's K1)	Gold Medalist